



<https://thevaultauth.vercel.app/>

THE VAULT

If they steal the password, the vault still stays shut.

WCS

2026 Hard Hacks @ UCSD





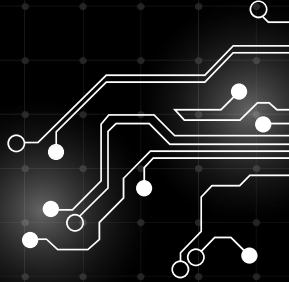
PROBLEM

The age of AI is ruining security.

Password breaks.

Deep fakes.

Even in your homes.



SOLUTION

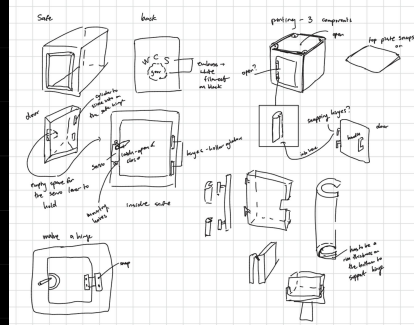
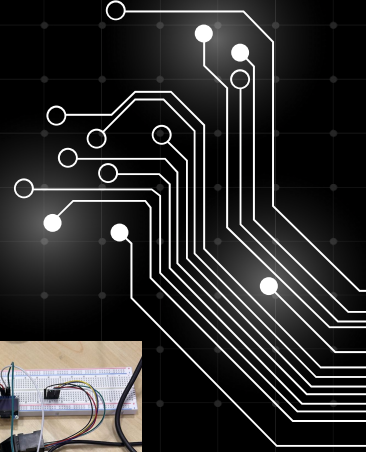
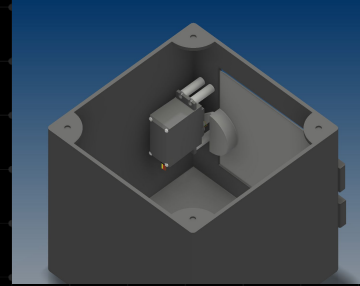
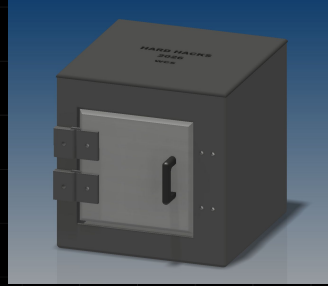
4-layer encrypted safe

- 4 digit keypad
- Fingerprint verification
- Camera
- LiDAR



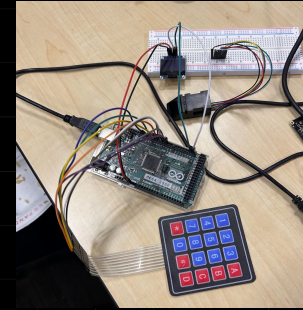
MECH

Sketching, Autodesk Inventor, 3D Printing



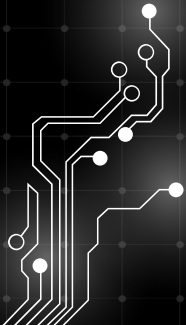
HARDWARE

Breadboarding, Rapid Prototyping



SOFTWARE

Backend, Frontend and Serial Communication

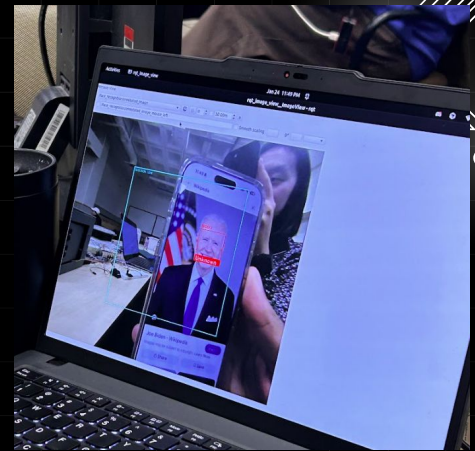


SOFTWARE

- ROS2 + Flask Backend
 - Face Recognition
 - Anti-spoofing
 - Database
- React Frontend
- PySerial for Arduino-Backend comms

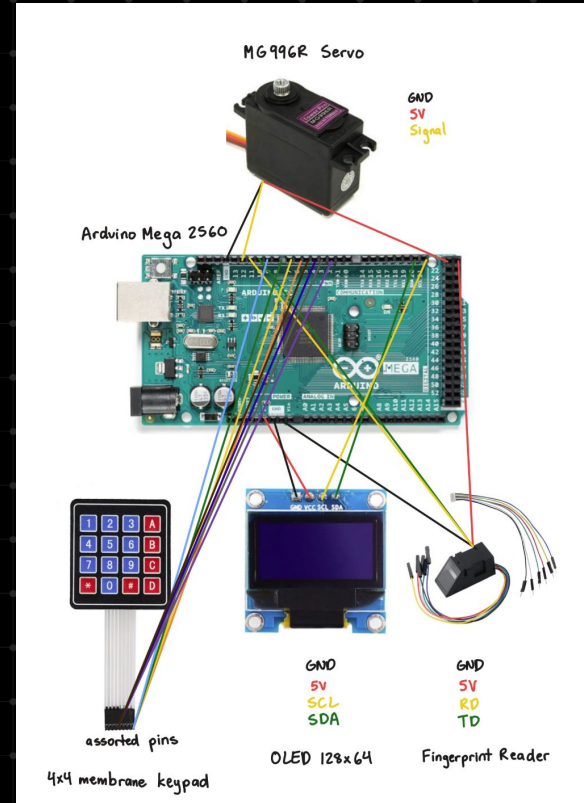
Pipeline:

Face Recognition -> Database Query -> Fingerprint
-> Hashed Keypad Password



HARDWARE

- Arduino Mega
- Intel NUC
- 4-layer system:
 - 4x4 membrane keypad
 - DIYMall Optical Fingerprint Reader Sensor Module
 - Oak-D Camera
 - Odin1 LiDAR
- SSD1306 OLED I2C 128x64
- Amazon Echo
- 24V Battery
- Buck Converter
- Centerboard
- Misc: breadboard, jumper wires





DEMO

